StockSense ESP32 Firmware Design Document

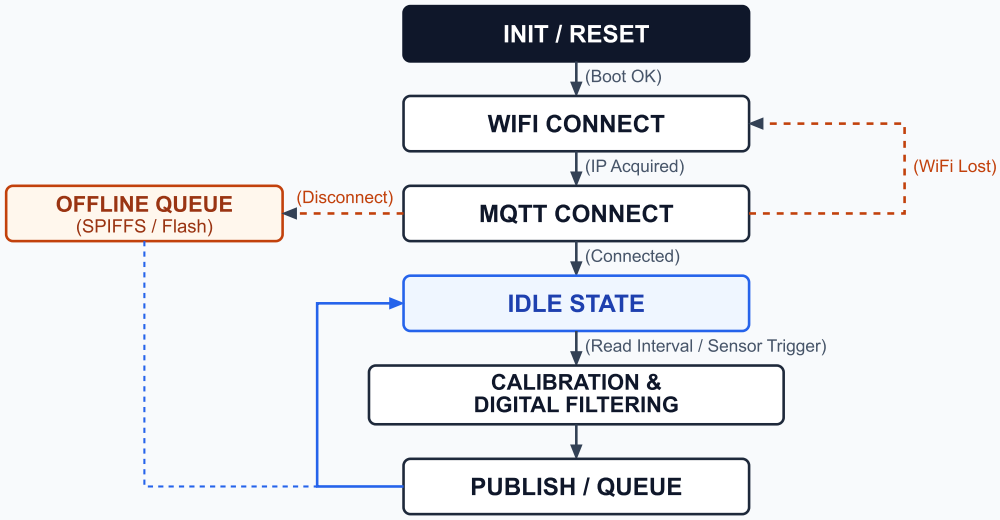
Project: StockSense Inventory Telemetry Node **Target MCU:** ESP32-WROOM-32 (DevKit)

1. FIRMWARE EXECUTIVE OVERVIEW & ARCHITECTURE

The StockSense firmware runs on the ESP32 dual-core microcontroller and provides real-time mass measurements (via 24-bit HX711 ADC), RFID item verification (RC522), and robust cloud telemetry synchronization over Wi-Fi/MQTT. Core design goals prioritize zero telemetry loss during transient network partitions, smooth low-noise sensor reading, and secure Over-The-Air (OTA) update mechanisms.

2. FINITE STATE MACHINE (FSM) ARCHITECTURE

The firmware is structured around an asynchronous, non-blocking finite state machine executing on FreeRTOS Core 1 as illustrated below.



State	Description & Action	Transition Trigger
STATE_INIT	Initializes LittleFS, GPIO pins, HX711 ADC, and RC522 SPI bus. Loads calibration parameters from Flash.	Hardware check OK → **STATE_WIFI_CONNECT**
STATE_WIFI_CONNECT	Attempts non-blocking Wi-Fi station connection. Times out after 10 seconds.	IP acquired → **STATE_MQTT_CONNECT** Timeout → **STATE_OFFLINE_MODE**
STATE_MQTT_CONNECT	Establishes TLS/TCP session with Mosquitto MQTT broker and subscribes to command topics.	MQTT Connected → **STATE_IDLE_RUNNING** Connection Failed → **STATE_OFFLINE_MODE**
STATE_IDLE_RUNNING	Polls sensor task timer (10 Hz), checks RFID presence, and processes incoming MQTT command frames.	Timer elapsed → **STATE_SAMPLE_FILTER** Network dropped → **STATE_OFFLINE_MODE**

State	Description & Action	Transition Trigger
STATE_SAMPLE_FILTER	Reads raw 24-bit HX711 counts, applies digital filtering (EMA), and calculates calibrated mass.	Sample completed → **STATE_PUBLISH_TELEMETRY**
STATE_PUBLISH_TELEMETRY	Serializes payload to JSON. Transmits via MQTT if connected; buffers to Flash if offline.	Flush offline queue → **STATE_IDLE_RUNNING**
STATE_OFFLINE_MODE	Appends telemetry packets to LittleFS circular buffer (`queue.bin`) and schedules retry timers.	Wi-Fi re-established → **STATE_WIFI_CONNECT**
STATE_OTA_UPDATE	Downloads firmware binary via HTTPS, verifies SHA-256 hash, and performs atomic partition swap.	Flash complete → Reboot Verification Error → **STATE_IDLE_RUNNING**

3. CALIBRATION & DIGITAL SIGNAL FILTERING

Raw load cell signals suffer from ambient vibration and electrical interference. The firmware processes raw 24-bit integer values from the HX711 using a zero-point Tare Offset followed by an **Exponential Moving Average (EMA)** low-pass filter:

> $S_t = \alpha \times X_t + (1 - \alpha) \times S_{t-1}$
> Where X_t is raw calibrated weight input, S_t is filtered output weight (grams), and smoothing coefficient $\alpha = 0.20$.

Calibration Equation: Mass (g) = (Raw ADC Count − Tare Offset) / Scale Factor.

4. OFFLINE QUEUE & FLASH PERSISTENCE

When network disconnects occur, the firmware writes JSON telemetry records into a binary ring buffer on LittleFS storage (`/spiffs/offline.dat`). Upon reconnecting to MQTT:

- The system drains buffered records sequentially at 50 ms intervals to avoid broker congestion.
- Ring buffer size is capped at 2.5 MB (~15,000 telemetry readings), ensuring up to 48 hours of continuous offline operation.

5. OVER-THE-AIR (OTA) ARCHITECTURE

OTA updates execute asynchronously on FreeRTOS Core 0 using standard ESP32 dual-partition layout (`ota_0` / `ota_1`). Firmware manifest instructions are received on `stocksense/device/{device_id}/ota`. SHA-256 signature verification guarantees integrity before writing to the secondary partition.